

Abstract Algebra I

Exercise Sheet 3: Bijections and Homomorphisms

Bijections

Recall that a function $f : X \rightarrow Y$ is *injective* if every two elements of the domain X are mapped to distinct elements of the range.

Example.

- $f : \mathbb{R} \rightarrow \mathbb{R} : x \mapsto 0$ is not injective as $f(1) = 0$ and $f(2) = 0$.
- $f : \mathbb{R} \rightarrow \mathbb{R} : x \mapsto x$ is injective as $f(x) = f(y)$ if and only if $x = y$, i.e. two numbers x and y are mapped to distinct numbers if and only if x and y are themselves distinct.

Recall that a function $f : X \rightarrow Y$ is *surjective* if every element of the range Y is mapped onto by some value of the domain.

Example.

- $f : \mathbb{R} \rightarrow \mathbb{R} : x \mapsto 0$ is not injective as no element maps to 1, i.e. $f(x) \neq 1$ for any $x \in \mathbb{R}$.
- $f : \mathbb{R} \rightarrow \mathbb{R} : x \mapsto 2x$ is surjective as for every element y in $\mathbb{R}$ there is a value that is mapped to y by f , in particular $f(y/2) = y$.

One can check for surjectivity of a function by computing its image.

Theorem. A function $f : X \rightarrow Y$ is surjective if and only if $\text{Im}(f) = Y$.

Question 1. Decide whether the following functions are injective, surjective, and bijective. If yes, give a proof, if no, give a counterexample.

- $f : \mathbb{R} \rightarrow \mathbb{R} : x \mapsto x^2$
- $f : \mathbb{R} \rightarrow \mathbb{R}_{\geq 0} : x \mapsto x^2$
- $f : \mathbb{R} \rightarrow \mathbb{R} : x \mapsto e^x$
- $f : \mathbb{R} \rightarrow \mathbb{R}_{\geq 0} : x \mapsto e^x$
- $f : \mathbb{R} \rightarrow \mathbb{R}_{> 0} : x \mapsto e^x$
- $f : \mathbb{R} \rightarrow \mathbb{R} : x \mapsto \begin{cases} -1, & x < 0, \\ 0, & x = 0, \\ 1, & x > 0. \end{cases}$
- $f : \mathbb{Z} \rightarrow \mathbb{Z} : x \mapsto 2x$
- $f : \mathbb{Z} \rightarrow \mathbb{Z} : x \mapsto x + 1$
- $f : \mathbb{Z} \rightarrow 2\mathbb{Z} : x \mapsto 2x$
- $f : \mathbb{Z} \rightarrow \mathbb{N} : x \mapsto |x|$
- $f : \mathbb{N} \rightarrow \mathbb{N} : x \mapsto x + 1$
- $f : \mathbb{N} \rightarrow \mathbb{Z} : x \mapsto x$

Recall a homomorphism is a function between groups where the group operations may be computed before or after the function is applied, i.e. $f(ab) = f(a)f(b)$. Recall that $\mathbb{R}^\times$ is the group of units in $\mathbb{R}$, that is, $\mathbb{R}^\times = \mathbb{R} \setminus \{0\}$.

One can check for injectivity of a homomorphism by computing its kernel.

Theorem. *A homomorphism $f : X \rightarrow Y$ is injective if and only if $\ker(f) = \{e_X\}$.*

Question 2. Decide whether each of the following functions are homomorphisms. If yes, give a proof and determine what type of homomorphism they are (homomorphism, isomorphism, automorphism, etc.). If no, provide a counterexample. The group operation is assumed to be addition unless otherwise specified.

- $f : \mathbb{Z} \rightarrow \mathbb{Z} : x \mapsto 0$
- $f : \mathbb{Z} \rightarrow \mathbb{Z} : x \mapsto 1$
- $f : \mathbb{Z} \rightarrow \mathbb{Z} : x \mapsto x$
- $f : \mathbb{Z} \rightarrow \mathbb{Z} : x \mapsto 2x$
- $f : \mathbb{Z} \rightarrow \mathbb{Z} : x \mapsto x + 1$
- $f : \mathbb{Z} \rightarrow \mathbb{Z} : x \mapsto -x$
- $f : \mathbb{Z} \rightarrow 2\mathbb{Z} : x \mapsto 2x$
- $f : \mathbb{Z} \rightarrow 2\mathbb{Z} : x \mapsto 4x$
- $f : (\mathbb{R}, +) \rightarrow (\mathbb{R}_{>0}, \cdot) : x \mapsto e^x$
- Let $\mathcal{M}_n(k)$ denote $n \times n$ matrices with entries in k . Let M^+ denote the transpose of M .

$$f : (\{M \in \mathcal{M}_n(\mathbb{R}) \mid \det(M) \neq 0\}, \cdot) \rightarrow (\{M \in \mathcal{M}_n(\mathbb{R}) \mid \det(M) \neq 0\}, \cdot) : M \mapsto M^+.$$

- Assuming usual properties of determinants:

$$\det : (\{M \in \mathcal{M}_n(\mathbb{R}) \mid \det(M) \neq 0\}, \cdot) \rightarrow \mathbb{R}^\times$$

- Assuming usual properties of traces:

$$\text{trace} : (\{M \in \mathcal{M}_n(\mathbb{R}) \mid \det(M) \neq 0\}, \cdot) \rightarrow (\mathbb{R}, +)$$

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$$f : (\{M \in \mathcal{M}_n(\mathbb{Q}) \mid (M_{ij}) = \frac{z_{ij}}{n}, z_{ij} \in \mathbb{Z} \det(M) = 1\}, \cdot) \rightarrow (\{M \in \mathcal{M}_n(\mathbb{Z})\}, +) \\ : (M)_{ij} \mapsto z_{ij}$$

Question 3. Let $\varphi : G_1 \rightarrow G_2$ be an isomorphism. Prove that the inverse map $\varphi^{-1} : G_2 \rightarrow G_1$ is also an isomorphism.

Question 4. Find two distinct automorphisms of $(\mathbb{Z}, +)$.

Question 5. Recall that $(\mathbb{Z}_n, +)$ is the group made up of $\{0, 1, \dots, n-1\}$ with addition defined modulo n , and that $(\mathbb{Z}_n, \cdot)$ is the group made up of elements from $\{1, \dots, n-1, n\}$ that are coprime to n with multiplication defined modulo n . Decide whether the two groups $(\mathbb{Z}_7, +)$ and $(\mathbb{Z}_7, \cdot)$ are isomorphic. Decide for which n the groups $(\mathbb{Z}_n, +)$ and $(\mathbb{Z}_n, \cdot)$ are isomorphic.

Question 6. Decide whether $(\mathbb{R}, +)$ and $(\mathbb{R}^\times, \cdot)$ are isomorphic.